Nicholas Kwan

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Education

University of Toronto Sept. 2025 – Present
PhD, Electrical Engineering *Toronto, ON, Canada*
- Supervisor: Prof. Wei Yu

University of Toronto Sept. 2023 – Jan. 2026
MASc, Electrical Engineering – 3.90/4.00 *Toronto, ON, Canada*
- Thesis: *Coded Acknowledgements: Random Subspaces for Random Access*
- Supervisor: Prof. Wei Yu

University of British Columbia Sept. 2017 – May 2023
BASc, Engineering Physics, Minor Hons. Math *Vancouver, BC, Canada*
- Thesis: Constructions for Nonadaptive Tropical Group Testing
- Supervisor: Prof. Lele Wang

Research Experience

Graduate Research Assistant Sept. 2023 – Present
U of T Electrical Engineering, Supervisor: Prof. Wei Yu *Toronto, ON, Canada*
- Constructed efficient scheme for acknowledgement in massive random access using random subspaces.
- Characterized information-theoretic bounds of acknowledgement in massive random access.

NSERC USRA Undergraduate Research Assistant May 2023 – Sept. 2023
UBC ECE, Supervisor: Prof. Lele Wang *Vancouver, BC, Canada*
- Project: High Dimensional Expanders
- Surveyed recent results on high dimensional expanders and applications to long-standing problems in coding and sampling. Presented material weekly to reading group.

Undergraduate Thesis Sept. 2022 – Jan. 2023
UBC ECE, Supervisor: Prof. Lele Wang *Vancouver, BC, Canada*
- Project: Tropical Group Testing
- Constructed random matrices over the tropical semiring that could be used for group PCR Testing.

NSERC USRA Undergraduate Research Assistant May 2020 – Sept. 2020
UBC ECE, Supervisor: Prof. Maryam Kamgarpour *Vancouver, BC, Canada*
- Project: Finding Nash Equilibria in Multi-agent Systems
- Tested algorithms to find Nash Equilibria in different classes of games.
- Found conditions for monotonicity in certain polymatrix games.

Publications

1. **Nicholas Kwan**, Ryan Song and Wei Yu, "Coded Acknowledgement with Random Subspaces" *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, China, July 2026.

2. Xuzhe Xia, **Nicholas Kwan** and Lele Wang, "On the Worst-Case Complexity of Gibbs Decoding for Reed–Muller Codes," *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, MI, USA, June 2025.

3. **Nicholas Kwan** and Lele Wang, "Constructions for nonadaptive tropical group testing," *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, Taipei, Taiwan, June 2023.

Work Experience

Graduate Teaching Assistant Sept. 2024 – Present
University of Toronto ECE *Toronto, ON, Canada*
- Tutorials and grading for ECE 367, Matrix Algebra and Optimization
- Office hours and grading for ECE 1505, Convex Optimization
- Office hours, grading, and writing exam problems for ECE 1502, Information Theory

Undergraduate Teaching Assistant Sept. 2019 – Dec. 2020
UBC Math, UBC Electrical and Computer Engineering *Vancouver, BC, Canada*
- TA for EECE 571U 2020W, Algorithmic Game Theory (Graduate Course).
- Facilitated workshops for MATH 100 2020W, Differential Calculus.
- Grader for MATH 184 2019W, Differential Calculus, and MATH 105 2019W, Integral Calculus.

Cryostat R&D Student Jan. 2019 – Apr. 2019
Center for Molecular and Materials Sciences, TRIUMF *Vancouver, BC, Canada*
- Designed, assembled, and tested cryostat inserts for a gas flow cryostat.
- Modelled and improved temperature convergence time for temperatures from 1.8 K to 250 K.

Projects

Ducted UAV Capstone Sept. 2021 – May 2022
ENPH 479 Capstone
- Designed and assembled chassis and mechanical subsystems of a controllable ducted fan drone.
- Tested and achieved stability of drone under fixed directional input.

Silicon Photonics Capstone Sept. 2020 – May 2021
ENPH 459 Capstone
- Researched and evaluated efficient simulation methods for optimizing photonic crystal cavities.
- Wrote Python code for Guided Mode Expansion simulations for cavity structures.
- Wrote Jupyter notebooks documenting the use of Legume in optimizing photonic crystal cavities.

Ackermann Steering System Design Sept. 2018 – May 2019
UBC Supermileage
- Led a group to design and fabricate Ackermann steering system for a 2229 mpg prototype vehicle.
- Designed and fabricated shell components for above vehicle.